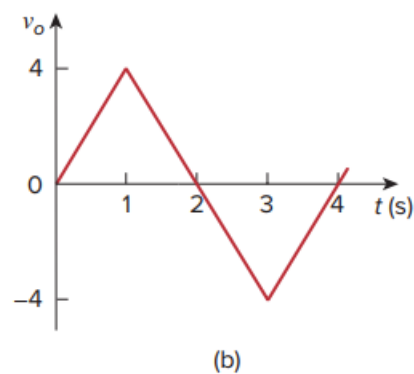
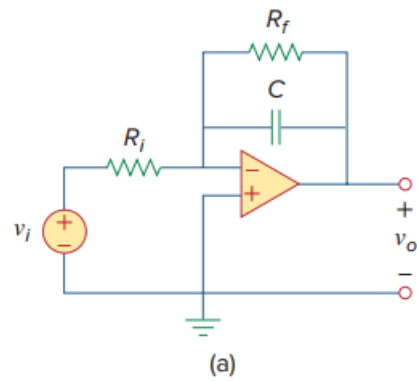


Problem

The output v_o of the op amp circuit in Fig. 6.92(a) is shown in Fig. 6.92(b). Let $R_i = R_f = 1\text{ M}\Omega$ and $C = 1\text{ }\mu\text{F}$. Determine the input voltage waveform and sketch it.

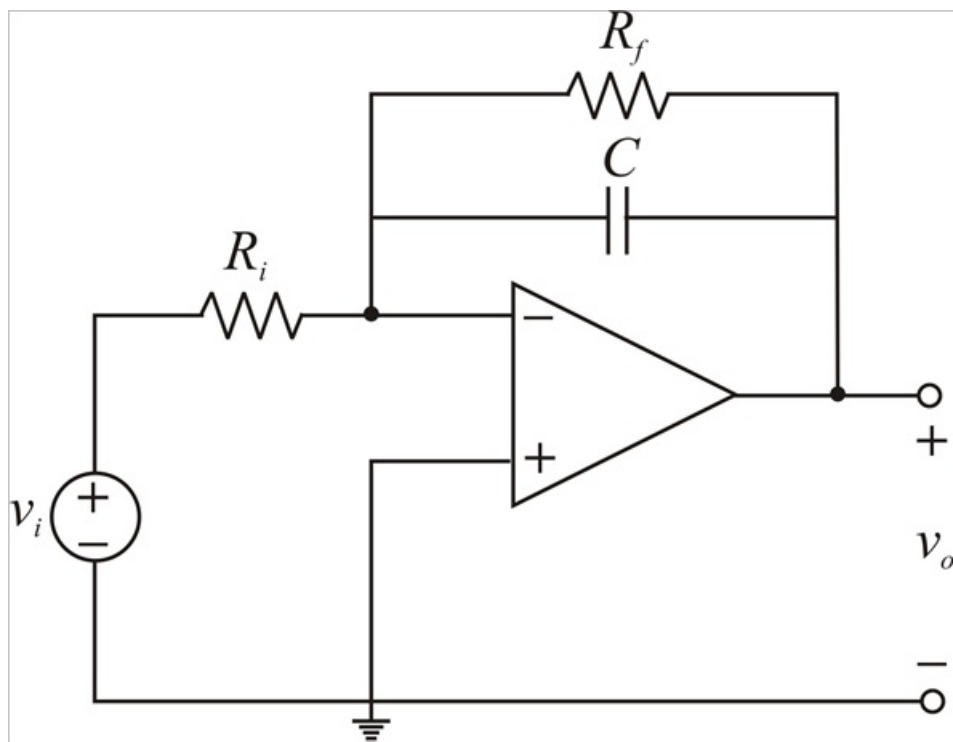
Fig. 6.92:



Step-by-step solution

Step 1 of 7

Given circuit diagram is



Step 2 of 7

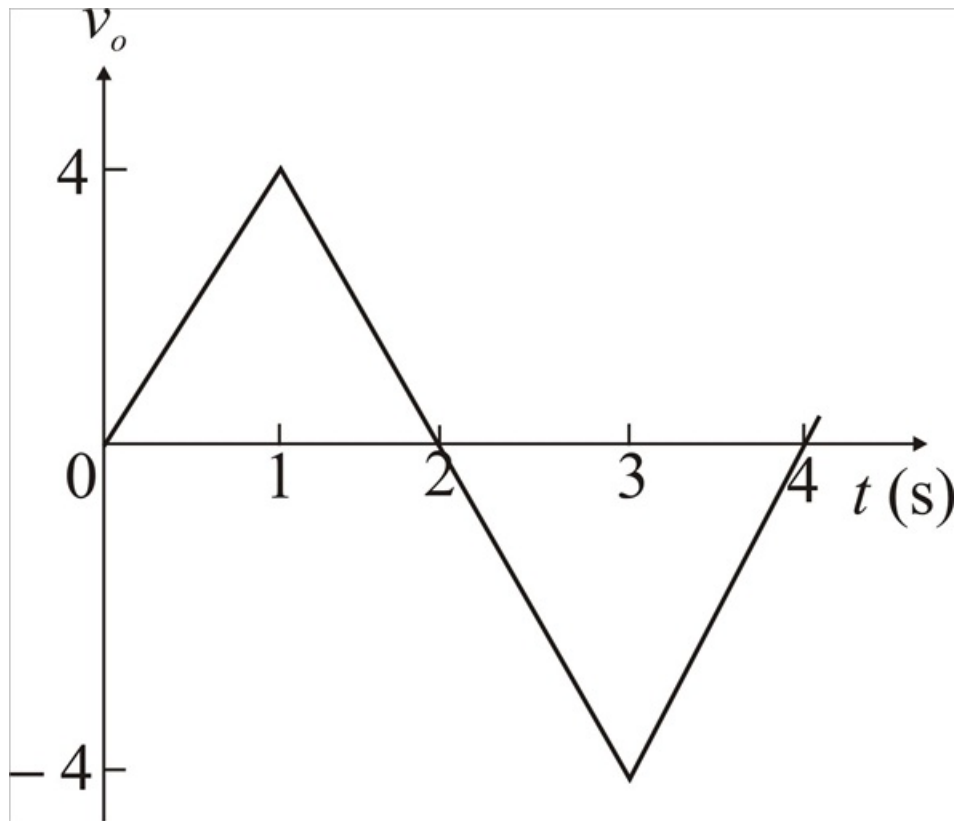
Given that

$$R_i = R_f = 1\text{ M}\Omega, \text{ and}$$

$$C = 1\text{ }\mu\text{F}.$$

Step 3 of 7

Given output wave form is



Step 4 of 7

From the above waveform, the output equation becomes

We know that, the slope equation

$$(y - y_1) = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$$

Here, $y = v_o$ and $x = t$

For $0 < t < 1\text{ s}$, we have $(0, 0)$ and $(1, 4)$ coordinates, then the slope equation is

$$(y - 0) = \frac{4 - 0}{1 - 0} (x - 0)$$

$$y = 4(x)$$

$$v_o = 4t.$$

For $1 < t < 3\text{ s}$, we have $(1, 4)$ and $(3, -4)$ coordinates, then the slope equation is

$$(y - 4) = \frac{-4 - 4}{3 - 1} (x - 1)$$

$$(y - 4) = \frac{-8}{2} (x - 1)$$

$$y - 4 = -4x + 4$$

$$v_o = -4t + 8.$$

For $3 < t < 4\text{ s}$, we have $(3, -4)$ and $(4, 0)$ coordinates, then the slope equation is

$$(y - (-4)) = \frac{0 - (-4)}{4 - 3} (x - 3)$$

$$(y + 4) = \frac{4}{1} (x - 3)$$

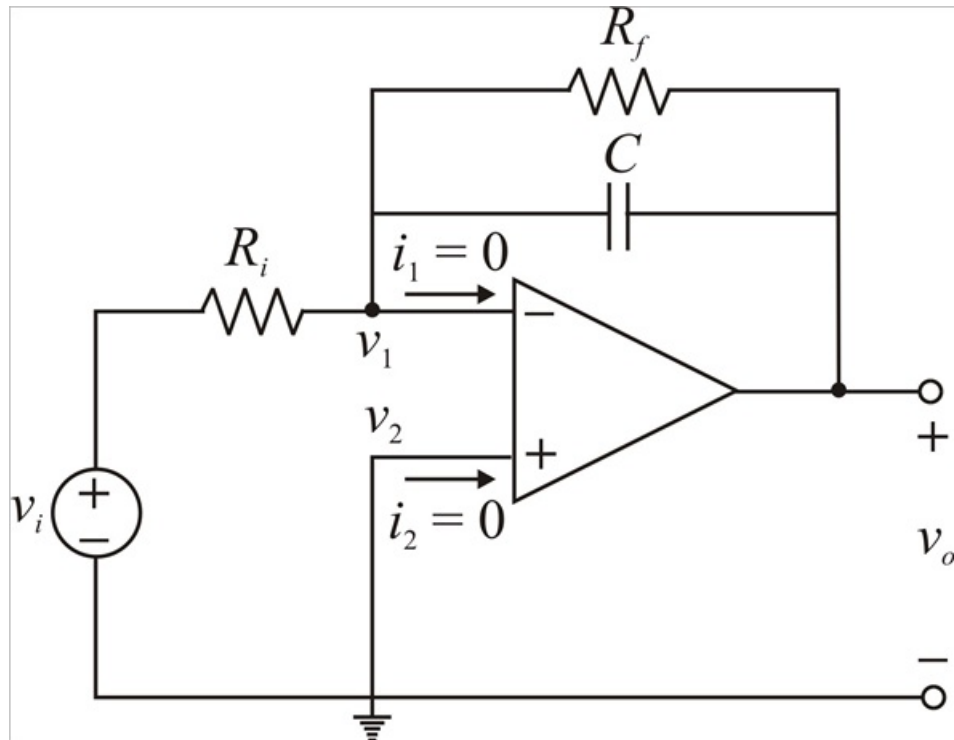
$$y + 4 = 4x - 12$$

$$v_o = 4t - 16.$$

Therefore, the output equation is

$$v_o = \begin{cases} 4t \text{ V} & \text{For } 0 < t < 1\text{ s} \\ -4t + 8 \text{ V} & \text{For } 1 < t < 3\text{ s} \\ 4t - 16 \text{ V} & \text{For } 3 < t < 4\text{ s} \end{cases}$$

Given circuit can be represented as below:



From the above circuit, $v_2 = 0 \text{ V}$.

In an ideal op-amp, the inverting and non-inverting terminal currents are zero, and the inverting and non-inverting node voltages are equal.

i.e.,

$$v_1 = v_2$$

$$\Rightarrow v_1 = v_2 = 0 \text{ V} \quad (\because v_2 = 0 \text{ V})$$

Applying KCL at node v_1 :

$$\frac{v_i - v_1}{R_i} = \frac{v_1 - v_o}{R_f} + C \frac{d(v_1 - v_o)}{dt}$$

$$\frac{v_i - 0}{R_i} = \frac{0 - v_o}{R_f} + C \frac{d(0 - v_o)}{dt} \quad (\because v_1 = v_2 = 0 \text{ V})$$

$$\frac{v_i}{R_i} = -\frac{v_o}{R_f} - C \frac{dv_o}{dt}$$

Substitute R_i, R_f and C in the above equation

$$\frac{v_i}{1 \times 10^6} = -\frac{v_o}{1 \times 10^6} - (1 \times 10^{-6}) \frac{dv_o}{dt}$$

$$\frac{v_i}{1 \times 10^6} = \frac{-v_o - (1 \times 10^6)(1 \times 10^{-6}) \frac{dv_o}{dt}}{1 \times 10^6}$$

$$v_i = -v_o - \frac{dv_o}{dt}$$

$$v_i = -\left(v_o + \frac{dv_o}{dt}\right) \text{ V}$$

Step 6 of 7

The input and output equations are

$$v_i = -\left(v_o + \frac{dv_o}{dt}\right) \text{ V} \dots\dots (1)$$

$$v_o = \begin{cases} 4t \text{ V} & \text{For } 0 < t < 1 \text{ s} \\ -4t + 8 \text{ V} & \text{For } 1 < t < 3 \text{ s} \dots\dots (2) \\ 4t - 16 \text{ V} & \text{For } 3 < t < 4 \text{ s} \end{cases}$$

Step 7 of 7

From equations-(1) and (2)

For $0 < t < 1\text{ s}$, the input equation becomes

$$v_i = -\left(4t + \frac{d(4t)}{dt}\right)$$

$$v_i = -(4t + 4) \text{ V}$$

For $1 < t < 3\text{ s}$, the input equation becomes

$$v_i = -\left(-4t + 8 + \frac{d(-4t + 8)}{dt}\right)$$

$$v_i = -(-4t + 8 - 4)$$

$$v_i = -(-4t + 4)$$

For $3 < t < 4\text{ s}$, the input equation becomes

$$v_i = -\left(4t - 16 + \frac{d(4t - 16)}{dt}\right)$$

$$v_i = -(4t - 16 + 4)$$

$$v_i = -(4t - 12)$$

Therefore, the input equation is

$$v_i = \begin{cases} -(4t + 4) \text{ V} & \text{For } 0 < t < 1\text{ s} \\ -(-4t + 4) \text{ V} & \text{For } 1 < t < 3\text{ s} \\ -(4t - 12) \text{ V} & \text{For } 3 < t < 4\text{ s} \end{cases}$$

From the above input equation, the waveform becomes

